3. **Vulnerability scanning:** This involves a process wherein a vulnerability scanner periodically scans the network for different weaknesses that open up the potential for an exploit. This network monitoring methodology is different from intrusion detection since it detects a weakness before the actual attack has taken place, whereas intrusion detection identifies various unauthorised entries after the hacker breaches the network.
4. **Penetration testing**: This is usually carried out by IT professionals by implementing methods that are used by hackers to breach a network. Such tests satisfy clients that the network can handle all the techniques used by hackers and that their network will not allow hackers to enter it. The ultimate purpose of this type of testing is to take the network security to another level by discovering various vulnerabilities that hackers may be aware of but have not yet been detected by other monitoring methods.
5. **Firewall monitoring:** Firewalls monitor the traffic that's coming in and going out of the network. They track all the activities of the firewall to ensure that the screening process for incoming and outgoing connections is working properly and securely.

Different open source tools used for network monitoring

Network monitoring tools eliminate the requirement for a physical systems administrator; organisations implementing these tools can save a lot of time and money. Let's take a quick look at three such tools.

TeemIP

TeemIP is basically a change management database system that combines the IP address management system with a trouble-ticketing system, so that different network devices and IP addresses can be managed in the context of organisations, locations, users and their roles. It also tracks change requests and user trouble.

Features

1. It's a Web application that runs on any AMP stack (e.g., Apache/IIS with PHP 5.3.6+ and MySQL 5.5.3+), on Linux, Windows, MacOS and Solaris, with all of the major browsers.
2. It can handle IPv4 and IPv6 address registrations, range planning and subnet. It supports capacity tracking and management with support for nesting as well, in order to allow delegation of IP spaces.
3. It has got the ability to integrate different external data sources like device discovery, and can import a huge set of data from CSV files. We can also export data to CSV, XML and HTML formats using Object Query Language.
4. It has an integrated change-ticketing and troubleshooting system. We can define ticketing-system users to be configuration managers, administrators, document authors, portal power users, helpdesk agents, or even a combination of all these roles.

Advantages

1. It has high scalability.
2. It provides consistent and comprehensive documentation of our network IP resources.
3. As its open source, it is available free of cost.

Node-RED

Node-RED is another open source network monitoring tool that is developed by IBM. It is basically a flow-based programming system that monitors different networks.

Features

1. It is based on the Node.js JavaScript. It runs on every OS that Node.js supports, which includes Linux, Windows, MacOS, AIX and SunOS. We can even run Node-RED on single-board computers like Raspberry Pi and Beaglebone with full support for all on-board input/output facilities. Now, it comes built-in to the Raspberry Pi's Raspbian OS.
2. Node-RED instances are being offered by a couple of cloud services including IBM Bluemix, SenseTecnic FRED, Amazon Web Services and Microsoft Azure.
3. It is a useful general-purpose application platform providing ad hoc and quick solutions for network monitoring. This makes it an invaluable addition to our digital toolkit.
4. It is completely browser-based, and uses the metaphor of wiring different nodes together.
5. Node-RED is available with many built-in nodes that take care of social connections, general input and output, and utility based functions.
6. The Node-RED site comprises a library of user contribution nodes, which currently include 817 flows and 1,360 nodes.
7. Node-RED has a dashboard that can help us create user interfaces with graphs, switches, sliders, buttons and so on.

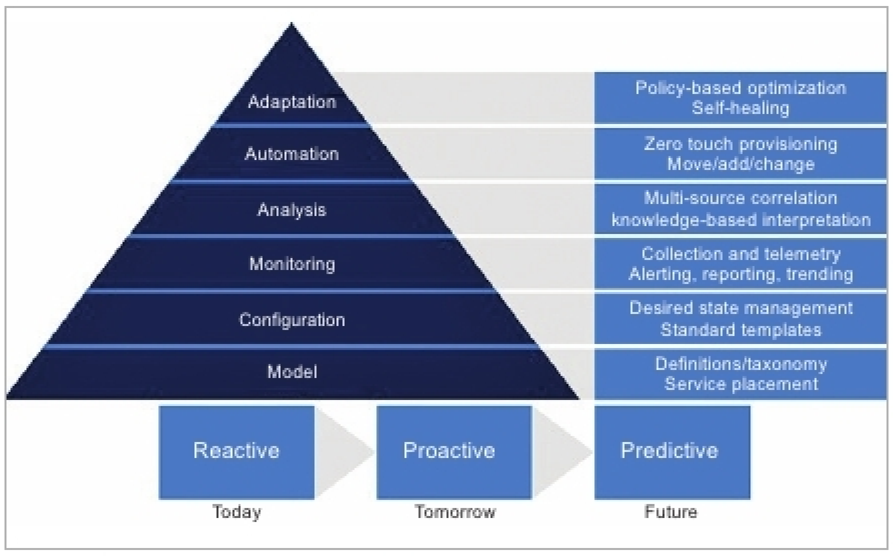

Figure 3: Pyramid diagram for a software monitored network (Image source: *googleimages.com*)